



Maximum sub-array-sum

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Given an array of n integers $x_1, x_2, \dots, x_n$, our task is to find the maximum sub array sum, i.e., For example, in the array $[-1, 2, 4, -3, 5, 2, -5, 2]$ the following subarray produces the maximum sum 10: $[-1, 2, 4, -3, 5, 2, -5, 2]$

Input Format

Two line input First line should read an integer value which represents size of array Second line consists list of elements(integers) separated by space

Constraints

1

Output Format

It prints integer value which represents maximum sub array sum

Sample Input 0

```
8
-1 2 4 -3 5 2 -5 2
```

Sample Output 0

```
10
```

Sample Input 1

```
7
-3 4 -1 2 1 -4 3
```

Sample Output 1

```
6
```

Sample Input 2

```
8
-1 2 4 -3 5 2 -5 2
```

Sample Output 2

```
10
```

Sample Input 3

```
6
-1 2 3 4 5 -15
```



Sample Output 3

14

Sample Input 4

7
-3 4 -1 2 1 -4 3

Sample Output 4

6

Sample Input 5

5
1 2 3 4 -10

Sample Output 5

10

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Language

C++11



```
1  #include <cmath>
2  #include <cstdio>
3  #include <vector>
4  #include <iostream>
5  #include <algorithm>
6  using namespace std;
7
8
9  int main() {
10 int n;
11     cin >> n;
12
13     vector<int> arr(n);
14     for (int i = 0; i < n; i++) {
15         cin >> arr[i];
16     }
17
18     int currentSum = arr[0];
19     int maxSum = arr[0];
20
21     for (int i = 1; i < n; i++) {
22         currentSum = max(arr[i], currentSum + arr[i]);
23         maxSum = max(maxSum, currentSum);
24     }
25
26     cout << maxSum;
27     return 0;
```

Line: 29 Col: 1

Run Code

Submit Code

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